Junha Chun

Ph.D. Student | Robot Learning, World Models & Representation Learning
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RESEARCH PROFILE

Ph.D. student developing physically grounded representations for robust robot learning and efficient world-model planning. Research spans latent actions, visual generalization, and robot foundation models, with co-first-author papers at ICLR 2026 and CVPR 2026 and a co-authored paper at CoRL 2026. Hands-on experience with SO-101 and WidowX robot platforms.

EDUCATION

Seoul National University · Seoul, South Korea
Ph.D. in Electrical and Computer Engineering, in progress | Mar 2019 - Present
Advisor: Taesup Kim, Mar 2023 - Present; previously Sangkyun Cha, Mar 2019 - Feb 2023.
Current focus: physical AI and world models. Earlier focus: machine learning lifecycle management.

Seoul National University · Seoul, South Korea
B.S. in Electrical and Computer Engineering | Mar 2015 - Feb 2019

SELECTED RESEARCH CONTRIBUTIONS

- **Efficient world-model planning.** Co-first-author work on Sparse Imagination (ICLR 2026): planning with sparse visual tokens reduced average planning time by 40.5% at 50% token dropout, with 69.7% vs. 69.8% success across the reported simulation tasks.
- **View-robust robot learning.** Co-first-author work on VILA (CVPR 2026): learned latent actions grounded in physical dynamics to improve generalization to unseen camera viewpoints, evaluated in simulation and on a real robot.
- **Representation geometry for generalization.** First-author preprint on how representational priors in pretrained backbones relate to world action model robustness under visual variation; under review at ICLR 2027.

CONFERENCE PUBLICATIONS

An asterisk (*) denotes equal contribution.

Sparse Imagination for Efficient Visual World Model Planning
J. Chun*, Y. Jeong*, T. Kim. **ICLR 2026.**
Paper · Project · OpenReview

Learning to Act Robustly with View-Invariant Latent Actions
Y. Jeong*, **J. Chun***, T. Kim. **CVPR 2026.**
Paper · Project

PoLAR: Factorizing Extent and Mode in Latent Actions for Robot Policy Learning
Y. Jeong, J. Yu, M. Jo, **J. Chun**, T. Kim. **CoRL 2026.**
Paper · Project

PREPRINTS & OTHER PUBLICATIONS

Preprints under review

How Representational Priors Shape Generalization in World Action Models
J. Chun, Y. Jeong, T. Kim. arXiv:2610.29168, 2026. Under review at ICLR 2027.
Paper

Uncovering Vulnerability of Vision-Language-Action Models under Joint-Level Physical Faults
M. Jo*, T. Kwon*, **J. Chun**, Y. Jeong, T. Kim. arXiv:2606.10501, 2026. Under review at ICLR 2027.
Paper

Journal, workshop & earlier preprint

Object-centric world model for language-guided manipulation
Y. Jeong*, **J. Chun***, S. Cha, T. Kim. ICLR World Model Workshop, 2025.
Paper

Toward Generalized Visualization System for Various CNN Models: Module and Layer View
Y. Kim, **J. Chun**, S. Son, H. Park. IEEE Access, 2024.
Paper

Spatial Semantic Embedding Network: Fast 3D Instance Segmentation with Deep Metric Learning
D. Zhang, **J. Chun**, S. Cha, Y. Kim. arXiv:2007.03169, 2020.
Paper

RESEARCH & ENGINEERING EXPERIENCE

Open Platform Ecosystem for Innovation in Future Automotive Technology | 2020 - 2021
Designed and implemented database infrastructure, data statistics and quality reporting, and a de-identification system for the Korea Automotive Technology Institute open automotive data platform.

Undergraduate Research Intern, Seoul National University | 2018
Worked on language instruction parsing for RoboCup@Home 2018, Montreal, Canada. Advisor: Byoungtak Zhang.

GRANTS & HONORS

- **NVIDIA Academic Grant** | Jul - Dec 2026. Temporally-Structured World Action Models for Reliable Robotics Planning; compute support: 8 × H100 GPUs.
- **National Science & Technology Scholarship** | 2015 - 2019. Full-tuition scholarship.

TECHNICAL SKILLS

Deep learning: PyTorch, Lightning, Accelerate, TensorFlow.
Robotics: Hugging Face LeRobot, MuJoCo; hardware experience with SO-101 and Trossen Robotics WidowX.
Programming & systems: Python, C++, Linux, Git, Slurm.

TEACHING & ACADEMIC SERVICE

Teaching assistant, Seoul National University: Programming for Data Science (Spring 2026, Fall 2025); Big Data and Knowledge Management System II (Fall 2022); Data Science Seminar (Spring 2020); Electrical and Computer Engineering Seminar III (Spring 2019).
Reviewer: NeurIPS 2026; ICLR 2027.

LANGUAGES

Korean: native · English: professional proficiency · Japanese: intermediate (JLPT N2, 2023)